

Long-Term Effects of COVID-19 on Memory and Cognition: A Data-Driven Analysis

Nada HJIOUAI, Radya HANNAD, Zongo Sotissi Roland

Abstract

The COVID-19 pandemic has not only caused significant respiratory complications but has also led to long-term cognitive and neurological impairments, commonly referred to as "Long COVID." This study investigates the impact of COVID-19 severity on memory, reaction time, and fatigue using a data-driven approach. We analyzed a simulated dataset of 1000 patients with Long COVID, employing statistical techniques such as ANOVA, Principal Component Analysis (PCA), and K-means clustering. Our findings reveal that patients with severe COVID-19 experience significantly worse cognitive outcomes, including reduced memory scores, longer reaction times, and higher fatigue levels. The study highlights the importance of considering disease severity and gender in post-COVID cognitive rehabilitation strategies.

Introduction

The COVID-19 pandemic has brought to light not only acute respiratory issues but also long-term cognitive and neurological effects. Many patients report persistent cognitive impairments, such as memory deficits, attention disorders, and fatigue, even after recovering from the acute phase of the illness. These symptoms, collectively known as Long COVID, have raised concerns about their long-term impact on patients' quality of life and professional capabilities. This study aims to explore the cognitive effects of Long COVID, focusing on memory, reaction time, and fatigue, using a data-driven approach. By analyzing a simulated dataset inspired by real-world trends, we seek to identify key factors influencing cognitive outcomes and provide insights for better patient management and rehabilitation.

Related Works

Several studies have documented the cognitive and neurological impacts of COVID-19. The PHOSP-COVID study in the UK found that 78% of post-COVID patients experi-

enced persistent cognitive impairments. Research from Imperial College London revealed significant reductions in brain volume among COVID-19 survivors, while a study published in *The Lancet* suggested that Long COVID could accelerate brain aging by up to 10 years. These findings underscore the need for further research into the long-term cognitive effects of COVID-19, particularly in relation to disease severity and demographic factors such as age and gender.

Methodology

Problem Formulation

The primary objective of this study is to analyze the impact of COVID-19 severity on cognitive functions, specifically memory, reaction time, and fatigue. We aim to answer the following research questions:

1. How does COVID-19 severity affect memory scores?
2. What is the relationship between COVID-19 severity and reaction time?
3. How does fatigue vary with COVID-19 severity?

Dataset and Preprocessing

We used a simulated dataset of 1000 patients with Long COVID, which included variables such as age, gender, memory scores, reaction times, and fatigue levels. The dataset was preprocessed to handle missing values, normalize variables, and remove outliers. The preprocessing steps ensured the data's quality and relevance for subsequent analysis.

Data Description

The dataset contains the following variables:

- **Age:** The age of the patient (ordinal).
- **Gender:** The biological sex of the patient (nominal).
- **Memory Score:** A score out of 100 assessing short-term and long-term memory (continuous).
- **Reaction Time:** The average reaction time in milliseconds (continuous).
- **Fatigue Score:** A score from 0 to 10 measuring the patient's fatigue level (ordinal).
- **COVID-19 Severity:** A categorical variable indicating the severity of COVID-19 (mild, moderate, severe).

Preprocessing Steps

The following preprocessing steps were applied:

- **Handling Missing Values:** Missing values were either imputed using the mean or removed based on the percentage of missing data.
- **Normalization:** Variables were normalized to ensure comparability across different scales.
- **Outlier Removal:** Outliers were identified and removed using the interquartile range (IQR) method.

Algorithms and Methods

We employed several statistical techniques to analyze the data:

- **ANOVA:** To compare mean memory scores, reaction times, and fatigue levels across different COVID-19 severity groups.
- **Principal Component Analysis (PCA):** To reduce the dimensionality of the data and identify key factors influencing cognitive outcomes.
- **K-means Clustering:** To group patients based on their cognitive profiles and identify patterns related to COVID-19 severity and demographic factors.

ANOVA

ANOVA was used to test for significant differences in memory scores, reaction times, and fatigue levels across the three severity groups (mild, moderate, severe). The null hypothesis (H_0) was that there are no differences between the groups, while the alternative hypothesis (H_1) was that at least one group differs significantly.

PCA

PCA was applied to reduce the dimensionality of the dataset while retaining the maximum amount of information. The first two principal components were analyzed to identify the most influential variables.

K-means Clustering

K-means clustering was used to group patients into three clusters based on their cognitive profiles. The optimal number of clusters was determined using the elbow method.

Results

ANOVA Results

Our analysis revealed significant differences in cognitive outcomes based on COVID-19 severity:

- **Memory Scores:** Patients with severe COVID-19 had significantly lower memory scores compared to those with mild or moderate cases ($p < 0.05$).
- **Reaction Time:** Severe COVID-19 patients exhibited longer reaction times, indicating slower cognitive processing ($p < 0.05$).
- **Fatigue Levels:** Fatigue was significantly higher in severe COVID-19 cases compared to mild and moderate cases ($p < 0.05$).

PCA Results

PCA identified two main components:

- **PC1:** Explained 36.66% of the variance and was primarily influenced by memory scores and reaction times.
- **PC2:** Explained 22.40% of the variance and was influenced by age and COVID-19 severity.

K-means Clustering Results

K-means clustering revealed three distinct patient groups:

- **Cluster 1:** Patients with moderate COVID-19 severity, moderate memory scores, and moderate fatigue levels.
- **Cluster 2:** Patients with severe COVID-19 severity, low memory scores, and high fatigue levels.
- **Cluster 3:** Patients with mild COVID-19 severity, high memory scores, and low fatigue levels.

Discussion

The results highlight the significant impact of COVID-19 severity on cognitive functions. Patients with severe COVID-19 experience worse memory, slower reaction times, and higher fatigue levels, which could have long-term implications for their quality of life and

professional performance. Interestingly, age did not significantly influence cognitive outcomes, suggesting that disease severity is a more critical factor. Gender differences were also observed, with women in one cluster showing better cognitive resilience compared to men.

Implications for Rehabilitation

The findings suggest that rehabilitation programs should be tailored based on the severity of COVID-19. Patients with severe cases may require more intensive cognitive rehabilitation to address memory deficits and fatigue. Additionally, gender-specific interventions may be necessary to account for differences in cognitive resilience.

Future Work

Future research should focus on longitudinal studies to track cognitive recovery over time and explore the effectiveness of rehabilitation programs. Additionally, incorporating more detailed demographic and clinical data could provide deeper insights into the factors influencing cognitive outcomes in Long COVID patients. Further studies could also investigate the role of other variables, such as pre-existing conditions and vaccination status, in cognitive outcomes.

Conclusion

This study demonstrates that COVID-19 severity significantly impacts cognitive functions, particularly memory, reaction time, and fatigue. The findings underscore the need for targeted interventions to address the cognitive sequelae of Long COVID, especially for patients with severe cases. By leveraging data-driven approaches, we can better understand the long-term effects of COVID-19 and develop more effective strategies for patient care and rehabilitation.